SESSION ONE

AI ENGINEERING & STATUSLINE SETUP

SHOWCASE // HANDS-ON // FAQ

SESSION STRUCTURE

Showcase

Framing the discipline. Why environmental configuration matters more than individual prompts.

Hands-On

Step-by-step configuration of a custom PowerShell statusline for Claude Code.

FAQs

Troubleshooting token usage, script performance, and multi-platform git integration.

PROMPT VS. CONTEXT

Prompt Engineering

THE SHORT-TERM TURN

The instructions you type into the chat. It is what you **say** in the moment to get a specific result.

Context Engineering

THE LONG-TERM FOUNDATION

Structuring persistent info (`CLAUDE.md` , skills). It is what the agent **already knows** before you speak.

TARGETING AUTONOMY

LEVEL	NAME	REQUIREMENT	GOAL
2-3	Coding Assistant	Prompt fundamentals	Current status for most teams
4	Developer as PM	**Context Engineering**	Hand off specs, verify output
5	Dark Factory	**Spec Engineering**	Unambiguous autonomous builds

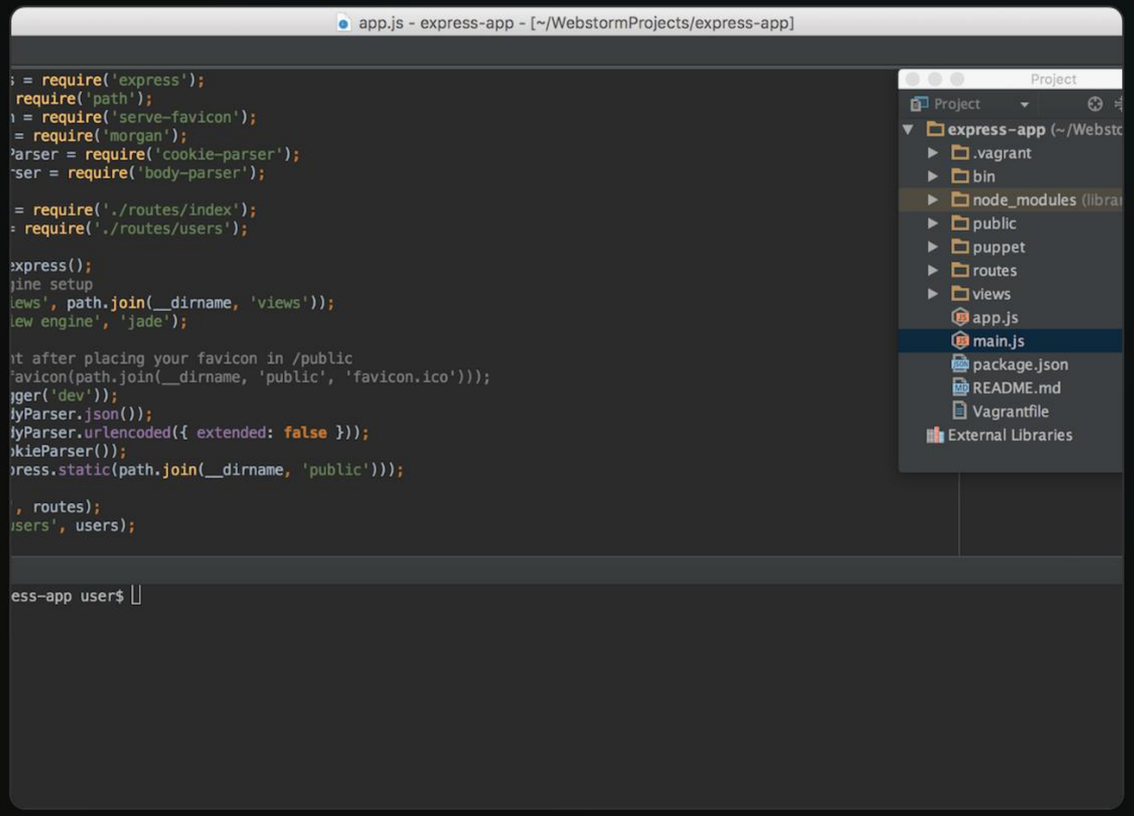
"Aim one level above where you are now."

WHY THE STATUSLINE?

Environment over Prompt

The statusline exercise demonstrates AI Engineering without writing app code:

- Low stakes, immediate feedback.
- Connects settings to data flow.
- Teaches agent environmental awareness.



MODULAR ARCHITECTURE

Entry Point

`statusline.ps1`

Reads stdin once and delegates to specialized scripts.

Basic Data

`statusline-basic.ps1`

Extracts model, folder, usage %, and session tokens.

Git Logic

`statusline-git.ps1`

Detects platform (GitHub/AzDO) and queries PR status.

JSON PRIMITIVES

DATA PIPED FROM CLAUDE CODE

FIELD	DESCRIPTION
`model.display_name`	Current active model (e.g. "Sonnet")
`context_window.used_percentage`	Current memory pressure (0-100)
`context_window.total_input_tokens`	Cumulative tokens sent to the API
`cost.total_duration_ms`	Total session wall-clock time

VISUALIZING CONTEXT

90%

Critical Threshold

When `used_percentage` hits 90%, the statusline bar shifts to **Bright Red**. This visual cue prompts the user to clear history or summarize context before the agent degrades.

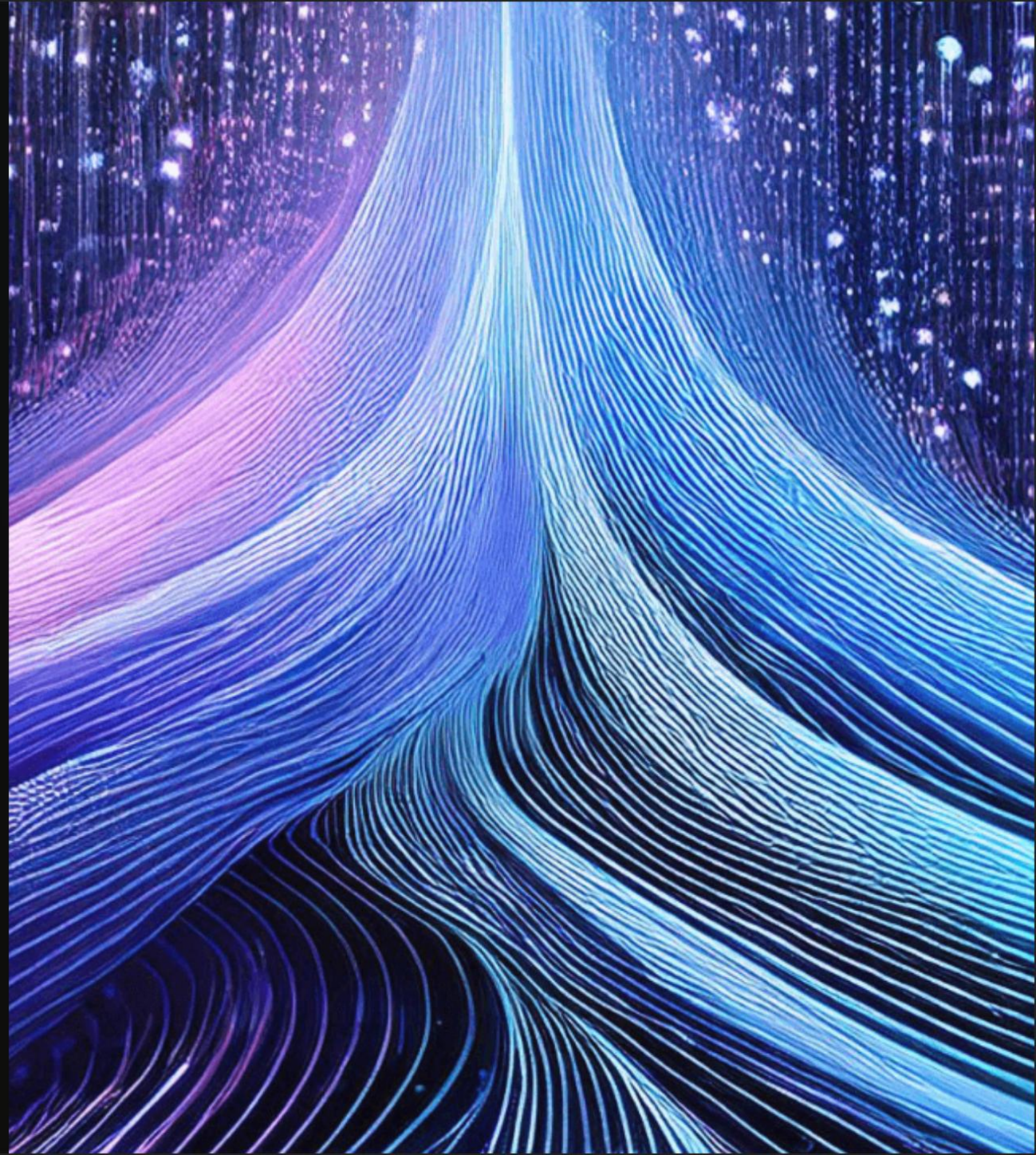
```
($usedPct -ge 90) { "$e[91m" }
```

PLATFORM INTEGRATION

GitHub & Azure DevOps

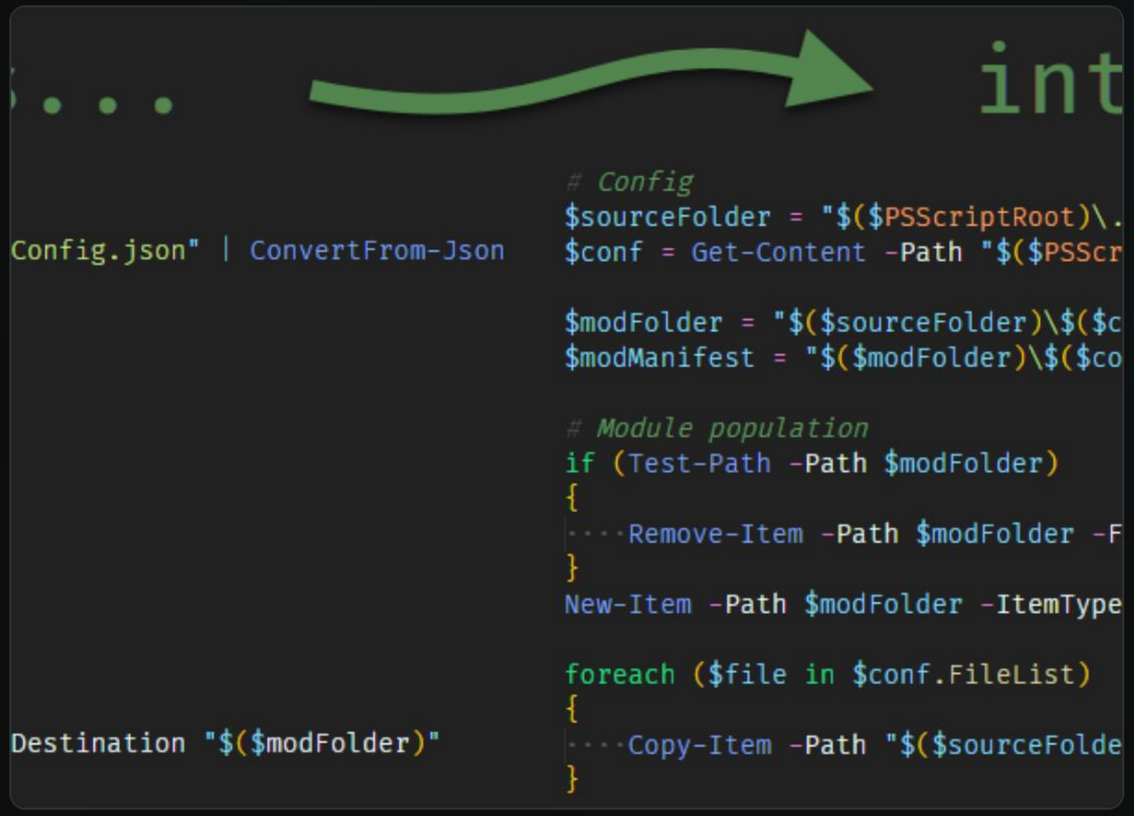
Using `gh` and `az` CLIs to pull live PR data into the terminal statusline.

Timeout logic (5s) prevents network latency from stalling the main agent interface.



CONFIGURATION SETTINGS

- **Location:** Edit `~/.claude/settings.json`
- **Command:** Use `pwsh -NoProfile` for faster startup.
- **Logging:** Debug script output in `debug.log`, not the terminal.
- **ANSI:** Use escape codes for colorization without app overhead.



COMMON PITFALLS

API Costs

The statusline runs **locally**. It does not consume API tokens or increase your monthly bill.

Slow Updates

Shelling out to CLIs is slow. Use `ProcessStartInfo` with timeouts to keep the UI snappy.

Garbled Text

Complex escape sequences clash. Stick to standard ANSI codes for the highest stability.

Questions?

Session One: Conclusion

Next Session: Advanced Context Engineering & CLAUDE.md mastery

IMAGE SOURCES

https://blog.jetbrains.com/wp-content/uploads/2016/01/webstorm-ws-mininalistic-layout.png

Source: blog.jetbrains.com

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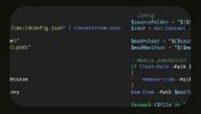
https://lumenalta.com/_next/image?url=https%3A%2F%2Fa-us.storyblok.com%2Ff%2F1019928%2F1920x728%2F68b5b790d3%2Fhero-banner-data-engineering-lifecycle.png%2Fm%2F&w=3840&q=75

Source: lumenalta.com



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Source: www.poshcodebear.com